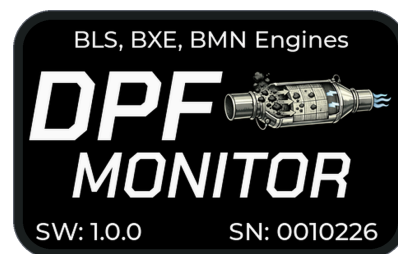


User Manual

DPF Load Monitor Mini

for VAG 1.9 / 2.0 TDI PD engines: **BLS, BXE, BMN**



BLS engine - good to know

- The ECU reports the distance since the last regeneration in **10 km steps** and the time since the last regeneration in **12-minute steps** - the values on the display change in these steps.
- According to VCDS data the **load coefficient** normally rises to about **40 %**, and then a regeneration starts.
- If the **carbon mass** is high (above approx. **50 g**) while the load coefficient stays low (around 10-15 %), the regeneration takes place about every **350 km**. This simply means the DPF is in good condition.

1 About the device

DPF Load Monitor Mini is a compact 1.64" AMOLED touch display that plugs into the OBD-II socket and reads live diesel particulate filter (DPF) data straight from the engine ECU: soot load, soot mass, DPF temperatures, differential pressure, distance and time since the last regeneration, boost pressure and more. It switches itself on with the engine and off when the engine stops, and beeps when a regeneration starts and ends.

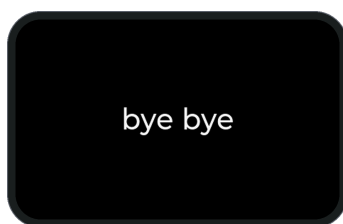
2 Installation and first start

- Start the engine first**, then plug the device into the vehicle's **OBD-II diagnostic socket** and route the cable so it does not obstruct the driver.
- With the engine running the welcome screen appears (engine family, software version **SW** and serial number **SN**), then the main screen with live values.
- When the engine is switched off the display shows *bye bye*, goes dark and the device goes to sleep. Nothing else is needed - it wakes up by itself at the next engine start.

⚠ Important: make the first connection to the OBD-II socket **with the engine running**. With the engine off the vehicle CAN bus is asleep and the device will not start.



Main screen



Engine stopped - going to sleep

3 Touch gestures

↔ **Swipe left / right**

Main screen: change page (6 pages). The thin bar at the bottom edge shows the current page.

↓ **Swipe down**

Main screen: open the **Menu** (from any page).

↑ **Swipe up**

Menu: back to the main screen.
Page 3: current → **MAX** → **MIN** → current. Page 6: reset MAX.

→ **Swipe right**

Settings and Info: back to the Menu.

● **Tap**

Main screen: toggle **Day / Night** brightness (only when Auto brightness is off). Menu: open an item.

⦿ **Hold 1 s**

Main screen: switch between **Standard** and **Big** layout (remembered after power-off). Keep the finger still.

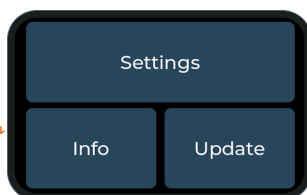
4 Screen map



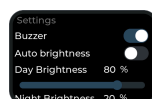
Main screen

6 pages

↕
swipe down +
swipe up ↕



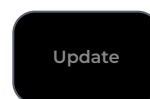
Menu



Settings
tap: Settings



Info
tap: Info



Update
hold ~2 s:
Update

↔ swipe right = back (Settings, Info → Menu)

5 Main screen - Standard layout

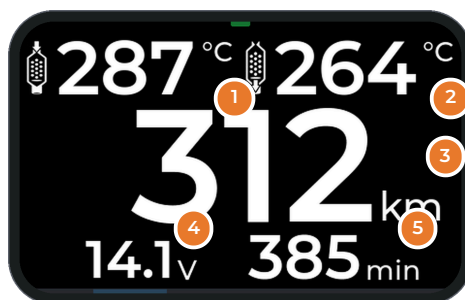
Swipe left/right to move between the six pages. Example values are shown.

Page 1 - DPF overview



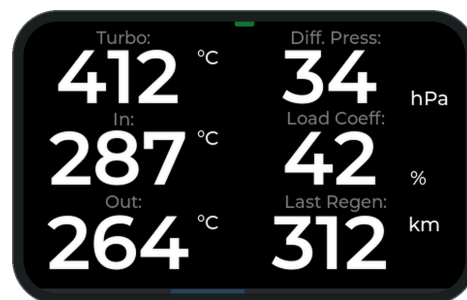
- 1 Load Coeff. - DPF load coefficient, %
- 2 Carbon Mass - soot mass in the DPF, g
- 3 DPF inlet temperature, °C
- 4 DPF outlet temperature, °C
- 5 distance since last regeneration, km
- 6 DPF differential pressure ΔP , hPa

Page 2 - Temperatures & regeneration



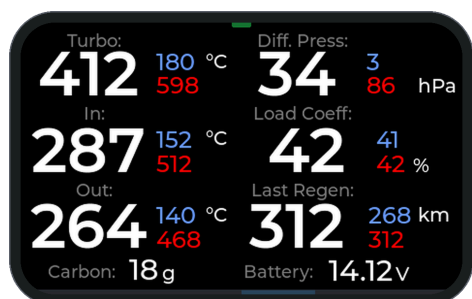
- 1 DPF inlet temperature, °C
- 2 DPF outlet temperature, °C
- 3 distance since last regeneration, km
- 4 battery (supply) voltage, V
- 5 time since last regeneration, min

Page 3 - Exhaust overview



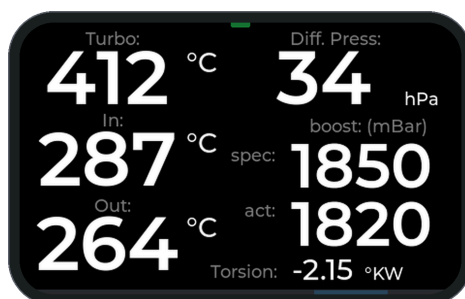
- Turbo - exhaust temperature at the turbocharger, °C
- In / Out - DPF inlet / outlet temperature, °C
- Diff. Press ΔP , Load Coeff. %, Last Regen km
- swipe up: MAX / MIN view (see below)

Page 4 - Minimum / maximum



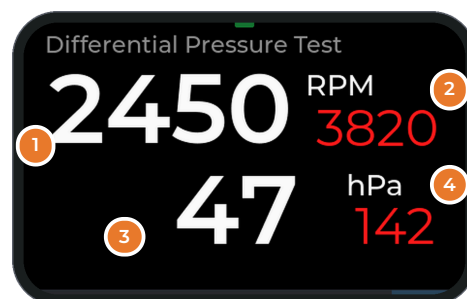
- Same values as page 3, plus the extremes recorded since the engine was started:
- red = maximum, blue = minimum
- bottom line: Carbon (soot mass, g) and Battery voltage

Page 5 - Engine



- Turbo, In, Out temperatures and ΔP
- boost spec / act - requested / actual boost pressure, mbar
- Torsion - camshaft timing (synchronisation) angle, °KW (crank degrees)

Page 6 - Differential pressure test



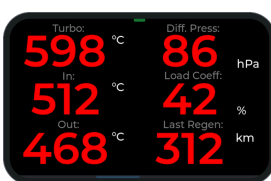
- 1 current engine speed, RPM
- 2 RPM at which the maximum ΔP was recorded
- 3 current DPF differential pressure, hPa
- 4 maximum ΔP since reset, hPa (swipe up = reset)

6 Page 3 - current / MAX / MIN view

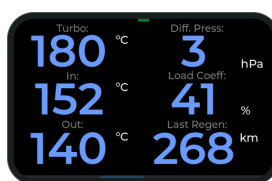
On page 3 swipe **up** to cycle the six values: **current** (white) → **MAX** (red) → **MIN** (blue) → current. Extremes are recorded from engine start. Changing the page returns to current values. Works in both layouts.



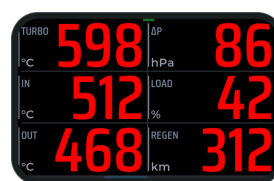
current



MAX



MIN



MAX - Big layout

7 Big layout

Hold your finger on the main screen for 1 s to switch to the Big layout - the same six pages with larger digits and short captions. Hold again to return to the Standard layout. The choice is remembered.



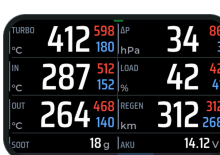
1



2



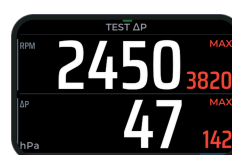
3



4



5



6

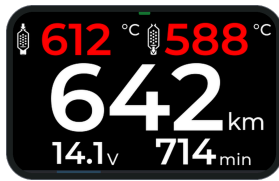
LOAD = load coeff. % · SOOT = soot mass g · IN / OUT = DPF inlet / outlet °C · REGEN = km since regeneration · ΔP = differential pressure hPa · CZAS = time since regeneration min · AKU = battery V · TURBO = turbo exhaust temp. °C · BOOST = actual / requested boost mbar · TORS. = camshaft timing angle °KW · MAX = maximum

8 DPF regeneration

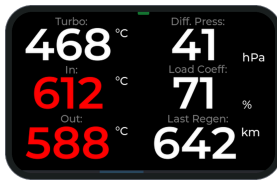
When the engine ECU starts a DPF regeneration, the device gives **two beeps** and shows **all DPF inlet and outlet temperatures in red** - on pages 1-5, in both the Standard and Big layout. Other values stay white. When the regeneration is finished it gives **one beep** and the temperatures turn white again. Beeps only with Buzzer switched on.



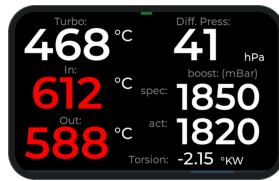
Page 1



Page 2



Page 3

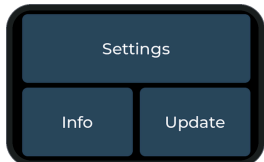


Page 5



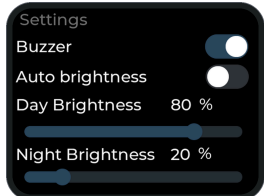
Big layout - page 1

9 Menu



- **Settings** - tap to open.
- **Info** - device information.
- **Update** - hold for about 2 s to restart the device in firmware-update mode (the screen goes dark). Use it only when installing new firmware according to the instructions supplied with the update.

10 Settings



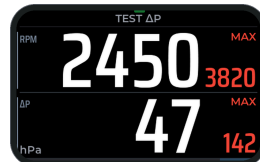
- **Buzzer** - sound signals on/off (regeneration start: 2 beeps, end: 1 beep).
- **Auto brightness** - the ambient-light sensor (on the top of the housing) switches between Day and Night brightness automatically. While on, tapping the main screen does not change brightness.
- **Day Brightness** (10-100 %) and **Night Brightness** (3-100 %) - the screen previews the level while you move the slider.
- All settings are saved automatically and kept after power-off. Scroll the screen up/down to see all items.

11 Info



Device name, hardware (**HW**) and software (**SW**) version, serial number (**SN**) and detected engine code. Scan the QR code to open dpfloadmonitor.pl.

12 Differential pressure (ΔP) test



1. Warm the engine up fully and go to page 6 (or its Big version).
2. Swipe up to reset the MAX values.
3. Only where it is safe and legal: in 3rd gear accelerate at full load up to about 4000 RPM.
4. Read **MAX ΔP** and the RPM at which it occurred. A DPF in good condition shows **below 200 hPa**. A clearly higher ΔP at the same RPM than in earlier tests indicates a clogged filter.

13 Automatic operation

The device switches on by itself after the engine is started and connects to the engine ECU. The small **green dot** at the top edge blinks while it is communicating with the ECU. After the engine is switched off it shows *bye bye*, the screen goes dark and the device goes to sleep. Min/max values are counted again from each engine start.

⚠ Do not operate the device while driving. Perform the ΔP test only on a closed or empty road, observing traffic rules.

DPF Load Monitor Mini · BLS / BXE / BMN · dpfloadmonitor.pl